Project Experience

RAG Intelligent Customer Service System

Project Description: A retrieval augmented generation customer service QA system combining external knowledge bases and LLMs to answer user queries.

Tech Stack: Python, FastAPI, LangChain, Elasticsearch, Kubernetes

Responsibilities:

1. Collected, cleaned, and standardized raw documents for ingestion into the knowledge base.

2. Split text, created embeddings, and stored vectors to enable semantic search capabilities.

3. Ran batch jobs on millions of historical QA records to evaluate accuracy, recall, and response time.

4. Wrote load testing and stability testing scripts to validate system performance.

Highlights:

1. Built a data access system supporting different knowledge base types.

2. Added resumable processing for very large batch tasks to prevent failure.

3. Improved QA accuracy with intent recognition, similar questions matching, and reranking.

RPA Workflow Automation Platform

Project Description: A platform utilizing software robots to automate repeated business operations across various applications.

Tech Stack: Python, Flask, SQLite, Docker, Linux

Responsibilities:

1. Developed modules for the designer, attended client, and unattended client interfaces.

2. Built reusable workflow components for email, Word, Excel, table, and database operations.

3. Managed client upgrade features and ensured backward compatibility during updates.

4. Utilized asynchronous libraries for efficient chunk upload and download operations.

Highlights:

1. Implemented scheduled robot task management using APScheduler.

2. Created reusable workflow components to reduce development time for new automations.

3. Supported unattended execution modes for high-volume background processing.

Operations Monitoring Platform

Project Description: A system monitoring network devices, servers, storage, and security devices to display collected metrics for users.

Tech Stack: Python, Django REST Framework, MySQL, RabbitMQ, Linux

Responsibilities:

1. Developed device report modules and self-monitoring functions for system health.

2. Investigated and fixed production problems to maintain high availability.

3. Supported distributed data collection and second-stage filtering before storage.

4. Implemented host-level monitoring for CPU, disk, connections, and MQ backlog indicators.

Highlights:

1. Added one-click export of monitoring metrics with visual charts.

2. Enabled real-time visibility into server and network device status.

3. Optimized data ingestion pipelines to handle high-frequency monitoring data.

Cloud Product Integration Deployment Script

Project Description: A script project handling conversion between standard products and cloud-integrated versions including installation and configuration.

Tech Stack: Python, MySQL, Linux, Git

Responsibilities:

1. Developed conversion scripts for installation, reverse proxy setup, and CMDB data push.

2. Performed self-testing, deployment, maintenance, and optimization of automation tools.

3. Used decorators to check whether files were changed before applying updates.

4. Wrapped configparser operations for centralized configuration management.

Highlights:

1. Automated service start and stop procedures during cloud integration.

2. Used threading and multiprocessing to replace frontend routes in batches.

3. Streamlined the deployment process reducing manual intervention requirements.